



Navigating Through the Test

During this test, you can use the mouse OR the keyboard to move through the screens.

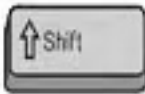
You can use the following standard keys on the keyboard:

Tab 

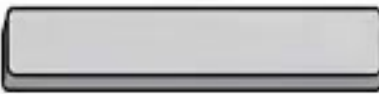
Press the Tab key to move from one selectable element on the page to the next. The current selectable element will be indicated by a thin blue border.

Examples of selectable elements:

- Buttons on the menu toolbar
- Answer options for test questions

Shift  and Tab 

Press the Shift and Tab keys to reverse direction through the selectable elements.

Spacebar 

Press the spacebar key to select an element. Press the spacebar key again to deselect the element.

Up  and Down 

Use the up and down arrows to scroll vertically.

Select **Continue** to proceed.



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Select **Continue** to proceed.



Test Center Regulations

You should have nothing on your computer table except your identification, pencil, and provided scratch paper. The official time will be kept by the computer. A test administrator is authorized to dismiss you from a test session and/or your scores may be withheld and ultimately canceled for any actions that violate the policies and procedures set forth herein and/or communicated at the test center including, but not limited to:

- Attempting to take the test for someone else or having someone else take the test for you
- Failing to provide acceptable identification
- Obtaining improper access to the test content, a part of the test or information about the test. This includes having test questions or answers in advance of the test administration and bringing preknowledge of test information into the test center in any form including, but not limited to, identification documents, prohibited devices and any other method listed as an aid in connection to the test.
- Using or having any prohibited device (e.g., phones, PDAs, all watches, including digital, analog and smartwatches, and other electronic, recording, listening, scanning or photographic devices) in your possession during the test administration, including during breaks
- Using any aids in connection with the test, including, without limitation: mechanical pencils, mechanical erasers, pens, pagers, beepers, all watches, including digital, analog and smartwatches, books, pamphlets, notes, unauthorized scratch paper, rulers, highlighter pens, scan pens or scanning devices, dictionaries, translators, compasses, protractors and any handheld electronic scanning, listening, recording or photographic devices
- Creating a disturbance or disruptive behavior in any form will not be tolerated. The test administrator has sole discretion in determining what constitutes disruptive behavior.
- Attempting to give or receive assistance or communication in any form is not permitted during the test administration. Discussion or sharing of test content or answers during the test administration, including breaks, and after the test administration, is prohibited.



If You Need the Administrator

If at any time during the test you think you have a problem with your computer or need the administrator for any reason, please raise your hand or otherwise indicate that you need the administrator.

Select **Continue** to proceed.



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I understand that, by selecting **Continue**, I will be given access to this confidential material only for the purpose of taking the test. I agree that I will not disclose to any person or entity, reproduce, or otherwise use the test questions, instructions, or other material revealed to me in the course of taking the test.

Select **Continue** to proceed.



Current Educational Status

Please indicate your current educational status:

- ☐ Sophomore (second year)
- ☐ Junior (third year)
- ☐ Senior (fourth or final year)
- ☐ First-Year Graduate Student
- ☐ Second-Year Graduate Student
- ☐ Unenrolled (College Graduate)
- ☐ Unenrolled (Master's Degree)
- ☐ Other

Intended Graduate Major Field

1. Find your intended Graduate Major Field.

Major Field Name:

Enter a full or partial name

Search



General Test Information

Timing Information

Total testing time on this test is 1 hour and 58 minutes.

If you wish to leave your seat during the test, please raise your hand or otherwise indicate that you need the administrator - timing will not stop.

Test Information

If you have a concern about the wording of a test question, please note the question number and continue the test. Report your concern to the administrator after you complete the test.

No credit will be given for any responses marked on scratch paper. Use the scratch paper to work out your answers. All scratch paper must be turned in to the administrator (or erased in front of the administrator in an at-home test) at the end of the testing session.

Within a timed section you may skip any question and return to it later before you exit the section. You should answer as many questions as possible during the test and manage your time with this in mind. Use the **Review** button at any time during the test to review which questions you have answered and which questions you have skipped.

Select **Continue** to proceed.

Analytical Writing

This section consists of a timed Analyze an Issue writing task. Standard timing for this section is 30 minutes. The timer displays the time remaining for the task.

Important Notice

Your essay response on the Analytical Writing section will be reviewed by ETS essay-similarity-detection software and by experienced essay readers during the scoring process. In light of the high value placed on independent intellectual activity within graduate schools and universities, your essay response should represent your original work. ETS reserves the right to cancel test scores of any test taker when an essay response includes, but is not limited to, any of the following:

- text that is unusually similar to that found in one or more other test essay responses;
- quoting or paraphrasing, without attribution, language that appears in published or unpublished sources; including sources from the Internet and/or sources provided by any third party;
- unacknowledged use of work that has been produced through collaboration with others without citation of the contribution of others;
- essays submitted as work of the test taker that appear to have been borrowed in whole or in part from elsewhere or prepared by another person.

When one or more of the preceding circumstances occurs, ETS may conclude, in its professional judgment, that the essay response does not reflect the independent writing skills that this test seeks to measure. When ETS reaches that conclusion, it cancels the Analytical Writing score, and because Analytical Writing scores are an integral part of the GRE General Test scores, those scores are canceled as well.

Select **Continue** to proceed.

In any field of endeavor, it is impossible to make a significant contribution without first being strongly influenced by past achievements within that field.

Write a response in which you discuss the extent to which you agree or disagree with the statement and explain your reasoning for the position you take. In developing and supporting your position, you should consider ways in which the statement might or might not hold true and explain how these considerations shape your position.

Cut

Paste

Undo

Redo

Confirm Early Exit on Analytical Writing Section

You still have time to write your response. As long as there is time remaining, you can keep writing or revising your response.

Select **Return** to keep writing or revising. Select **Continue** to leave this section.

Once you leave this section you **WILL NOT** be able to return to it.



Section 2 of 5

Verbal Reasoning

12 Questions

18 minutes (standard time)

For each question, indicate the best answer using the directions given. If you need more detailed directions, select **Help** at any time.

If a question has answer choices with **ovals** ☐, then the correct answer consists of a single choice. If a question has answer choices with **square boxes** ☐, then the correct answer consists of one or more answer choices. Read the directions for each question carefully. The directions will indicate if you should select one or more answer choices. To answer questions based on a reading passage, you may need to scroll to read the entire passage. You may also use your keyboard to navigate through the passage.

Select **Continue** to proceed.

Select one entry for the blank. Fill the blank in the way that best completes the text.

We pride ourselves on being the only species that has a concept of _____, yet we have a confounding habit of building barricades against perceived dangers while leaving ourselves exposed to real ones.

humility
risk
toleration
compassion
fear

Select one answer choice.

For each blank select one entry from the corresponding column of choices. Fill all blanks in the way that best completes the text.

It was not until 1995 that a planet beyond our solar system was first sighted, a discovery that greatly excited astronomers. Many had supposed that the processes that gave rise to our solar system were not (i) _____, and that there were other planets in the universe. Now, observations had (ii) _____.

Blank (i)

manifest
replicable
unique

Blank (ii)

caught up with belief
provided grounds for skepticism
put assumptions to the test

Select one entry from each column.

For each blank select one entry from the corresponding column of choices. Fill all blanks in the way that best completes the text.

Nineteenth-century England presents something of (i) _____, at least to those who believe that (ii) _____ best promotes business ethics. The defining characteristic of English company law at that time was its extreme degree of permissiveness: laissez-faire ruled in Parliament, caveat emptor in the courts, creating the ideal breeding ground for fraud and dishonesty. Yet these were the years when London became the financial center of the world and established its status as a (iii) _____ place to do business.

Blank (i)

a precedent
an enigma
an inspiration

Blank (ii)

innovation
individualism
regulation

Blank (iii)

prohibitively expensive
comparatively safe
culturally diverse

Select one entry from each column.

Questions 4 and 5 are based on this passage.

While a portrait of thirteenth-century Tibetan abbot Shangton Chögyi Lama observes contemporary iconographic rules for depicting an enlightened being, such as the temple-throne setting and gesture of religious instruction, the artist has also taken care to show details, such as lines around the mouth and a receding hairline, that suggest the portrayal of a unique individual. However, the painting's apparent realism is put into perspective when it is compared with a strikingly similar twelfth-century textile portrait. Similarities include not only body type and physiognomy but also details such as hairline and wrinkles. How closely these portraits resembled their subjects is unknown, but the close resemblance between the two portrayals of different men suggests the artists relied heavily on iconographic types.

Which of the following best describes the function of the highlighted phrase?

- ☐ It emphasizes the artist's skill and experience as a portraitist.
- ☐ It supports the author's assertion that Shangton Chögyi Lama was viewed as an enlightened being.
- ☐ It provides evidence for the portrait's apparent realism.
- ☐ It makes a connection between the portrait of Shangton Chögyi Lama and other portraits by the same artist.
- ☐ It puts the painting's apparent realism into perspective.

Select one answer choice.

Questions 4 and 5 are based on this passage.

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Consider each of the choices separately and select all that apply.

Which of the following statements about the textile portrait can be inferred from the passage?

- ☐ Its date precludes its being a copy of the portrait of Shangton Chögyi Lama.
- ☐ Its subject is portrayed as having a receding hairline.
- ☐ It presents a less detailed representation of its subject than does the portrait of Shangton Chögyi Lama.

Select one or more answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

Though Pauline Kael's long essays about film were popular in the 1960s and 1970s, ambitious professional arts criticism like hers is an increasingly _____ calling in a digital world where the old maxim "Everyone's a critic" is literally true.

- ☐ respectable
- ☐ embattled
- ☐ outmoded
- ☐ futile
- ☐ estimable
- ☐ obsolete

Select two answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

Well organized and researched and including all significant discoveries and medical scientists, this history of Western medicine has justly been called _____.

- ☐ encyclopedic
- ☐ long-winded
- ☐ exhaustive
- ☐ rambling
- ☐ overbearing
- ☐ indiscriminating

Select two answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

Though the art of writing was well established in Egypt and Mesopotamia long before the Shang period, the Chinese writing that appeared then seems to have been an entirely _____ development.

- ☐ unique
- ☐ banal
- ☐ homegrown
- ☐ practical
- ☐ routine
- ☐ indigenous

Select two answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

Neuroscientists are discovering that our sensory systems are much more interconnected than previously thought: even in ordinary circumstances, our senses can _____ one another in extraordinary ways.

- ☐ overrule
- ☐ bypass
- ☐ abet
- ☐ improve
- ☐ circumvent
- ☐ support

Select two answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

Neuroscientists are discovering that our sensory systems are much more interconnected than previously thought: even in ordinary circumstances, our senses can _____ one another in extraordinary ways.

- ☐ overrule
- ☐ bypass
- ☐ abet
- ☐ improve
- ☐ circumvent
- ☐ support

Select two answer choices.

Questions 10 to 12 are based on this passage.

Until the mid-1980s, Swedish workers' wages were largely determined through centralized collective bargaining at a national level, a policy aimed at achieving equal pay for equal work across companies and industries. This policy, designed to raise the relative wages of low-wage workers, resulted in a decrease in overall wage inequality in Sweden's labor market. Furthermore, this policy may have indirectly resulted in a relatively small gender wage gap (difference between women's wages and those of similarly qualified men): several recent studies have demonstrated that decentralized wage bargaining produces relatively large wage inequality in general and large gender wage differentials in particular, and international comparisons show that a society's overall wage inequality is positively related to the gender wage gap.

Therefore, in designing a study of organizational factors that perpetuate inequities in women's wages, researchers decided to gather data on the current situation in Swedish companies. The researchers reasoned that the occurrence of discriminatory wage-setting practices in Swedish companies could provide a conservative estimate of the prevalence of such practices in other developed countries. Despite increasing decentralization of Sweden's wage-bargaining process since the mid-1980s, they reasoned, the effects of the earlier policy on current wages should still be evident to some extent.

It can be inferred that the highlighted "studies" lend support to which of the following statements about women workers' wages?

- ☐ In most countries, women workers' wages would not likely be affected by centralized collective bargaining in the same way as they were in Sweden prior to the mid-1980s.
- ☐ In most countries, women workers' wages are closer to the wages of similarly qualified men than they are in Sweden.
- ☐ Women workers' wages are generally less affected by centralized collective bargaining than are men's wages.
- ☐ Women workers' wages are sometimes adversely affected by wage-setting practices that are intended to remedy overall wage inequality.
- ☐ Women workers' wages are generally closer to the wages of similarly qualified male workers where wages are set through centralized wage bargaining than where they are set through decentralized wage bargaining.

Select one answer choice.

Questions 10 to 12 are based on this passage.

Until the mid-1980s, Swedish workers' wages were largely determined through centralized collective bargaining at a national level, a policy aimed at achieving equal pay for equal work across companies and industries. This policy, designed to raise the relative wages of low-wage workers, resulted in a decrease in overall wage inequality in Sweden's labor market. Furthermore, this policy may have indirectly resulted in a relatively small gender wage gap (difference between women's wages and those of similarly qualified men): several recent studies have demonstrated that decentralized wage bargaining produces relatively large wage inequality in general and large gender wage differentials in particular, and international comparisons show that a society's overall wage inequality is positively related to the gender wage gap.

Therefore, in designing a study of organizational factors that perpetuate inequities in women's wages, researchers decided to gather data on the current situation in Swedish companies. The researchers reasoned that the occurrence of discriminatory wage-setting practices in Swedish companies could provide a conservative estimate of the prevalence of such practices in other developed countries. Despite increasing decentralization of Sweden's wage-bargaining process since the mid-1980s, they reasoned, the effects of the earlier policy on current wages should still be evident to some extent.

The author of the passage attributes the highlighted "decrease" to which of the following?

- ☐ A decline in Sweden's gender wage gap
- ☐ A decline in low-wage workers' wages in Sweden
- ☐ A decline in women's wages in Sweden
- ☐ A policy governing wage bargaining in Sweden
- ☐ A change in Sweden's wage-bargaining practices initiated in the mid-1980s

Select one answer choice.

Questions 10 to 12 are based on this passage.

Until the mid-1980s, Swedish workers' wages were largely determined through centralized collective bargaining at a national level, a policy aimed at achieving equal pay for equal work across companies and industries. This policy, designed to raise the relative wages of low-wage workers, resulted in a decrease in overall wage inequality in Sweden's labor market. Furthermore, this policy may have indirectly resulted in a relatively small gender wage gap (difference between women's wages and those of similarly qualified men): several recent studies have demonstrated that decentralized wage bargaining produces relatively large wage inequality in general and large gender wage differentials in particular, and international comparisons show that a society's overall wage inequality is positively related to the gender wage gap.

Therefore, in designing a study of organizational factors that perpetuate inequities in women's wages, researchers decided to gather data on the current situation in Swedish companies. The researchers reasoned that the occurrence of discriminatory wage-setting practices in Swedish companies could provide a conservative estimate of the prevalence of such practices in other developed countries. Despite increasing decentralization of Sweden's wage-bargaining process since the mid-1980s, they reasoned, the effects of the earlier policy on current wages should still be evident to some extent.

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Select one answer choice.

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The passage suggests which of the following about the highlighted "policy" ?

- ☐ It was abandoned in the mid-1980s because it was no longer considered effective.
- ☐ Its effect on Sweden's labor market has been debated in several recent studies.
- ☐ Its effect on Sweden's labor market was consistent with the intentions that led to its adoption.
- ☐ It produced an indirect result that ran counter to a widespread assumption about the effect of centralized collective bargaining on wages.
- ☐ It was modeled on policies that had been adopted in several other developed countries.

Select one answer choice.



End of Section

You have reached the end of this section. You have time remaining to review. As long as there is time remaining, you can check your work. Once you leave this section, you WILL NOT be able to return to it.

Select **Review** to go back to the Review screen.

Select **Return** to go to the last question in this section.

Select **Continue** to proceed to the next section of the test.



Section 3 of 5

Quantitative Reasoning

12 Questions

21 minutes (standard time)

For each question, indicate the best answer, using the directions given. If you need more detailed directions, select **Help** at any time.

An on-screen calculator is available for each question in this section. To use the calculator, select the calculator button.

If a question has answer choices with **ovals** ☐, then the correct answer consists of a single choice. If a question has answer choices with **square boxes** ☐, then the correct answer consists of one or more answer choices. Read the directions for each question carefully. The directions will indicate if you should select one or more answer choices. To answer questions based on a data presentation, you may need to scroll or use your keyboard to access the entire presentation.

All numbers used are real numbers.

All figures are assumed to lie in a plane unless otherwise indicated.

Geometric figures, such as lines, circles, triangles, and quadrilaterals, **are not necessarily** drawn to scale. That is, you should **not** assume that quantities such as lengths and angle measures are as they appear in a figure. You should assume, however, that lines shown as straight are actually straight, points on a line are in the order shown, and more generally, all geometric objects are in the relative positions shown. For questions with geometric figures, you should base your answers on geometric reasoning, not on estimating or comparing quantities by sight or by measurement.

Coordinate systems, such as xy -planes and number lines, are drawn to scale; therefore you can read, estimate, or compare quantities in such figures by sight or by measurement.

Graphical data presentations, such as bar graphs, circle graphs, and line graphs, **are** drawn to scale; therefore, you can read, estimate, or compare data values by sight or by measurement.

Select **Continue** to proceed.



x and y are positive consecutive odd integers.

$$xy < x + y$$

Quantity A

$$x + y$$

Quantity B

$$4$$

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.



$$\begin{aligned}x &> 1 \\ -1 &< y < 0\end{aligned}$$

Quantity A

$$x^{-2}$$

Quantity B

$$y^{-2}$$

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.



a , b , and c are constants.

$$b > 0$$

Quantity A

The y -intercept of the graph of the equation $ax + 64y = b$ in the xy -plane

Quantity B

The y -intercept of the graph of the equation $cx - 32y = -2b$ in the xy -plane

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.

Select one answer choice.



The average (arithmetic mean) of three test scores is 70. All of the test scores are between 30 and 80, inclusive.

Quantity A

The median of the three scores

Quantity B

75

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.



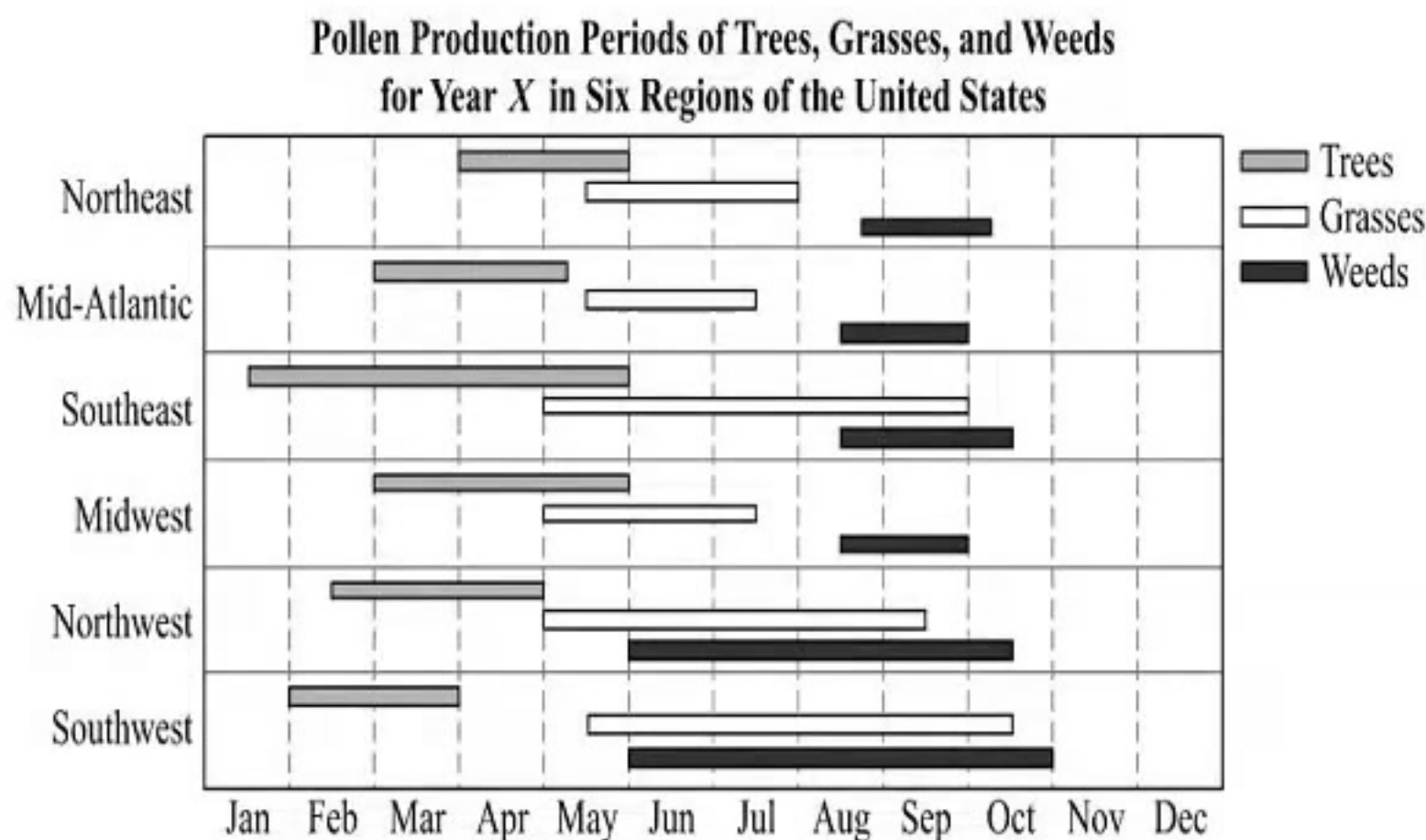
If the ratio of 700,000 to 12,500 is equal to the ratio of x to 20,000, what is the value of x ?

- ☐ 357
- ☐ 3,571
- ☐ 11,200
- ☐ 1,120,000
- ☐ 11,200,000

Select one answer choice.



Questions 6–8 are based on the following data.



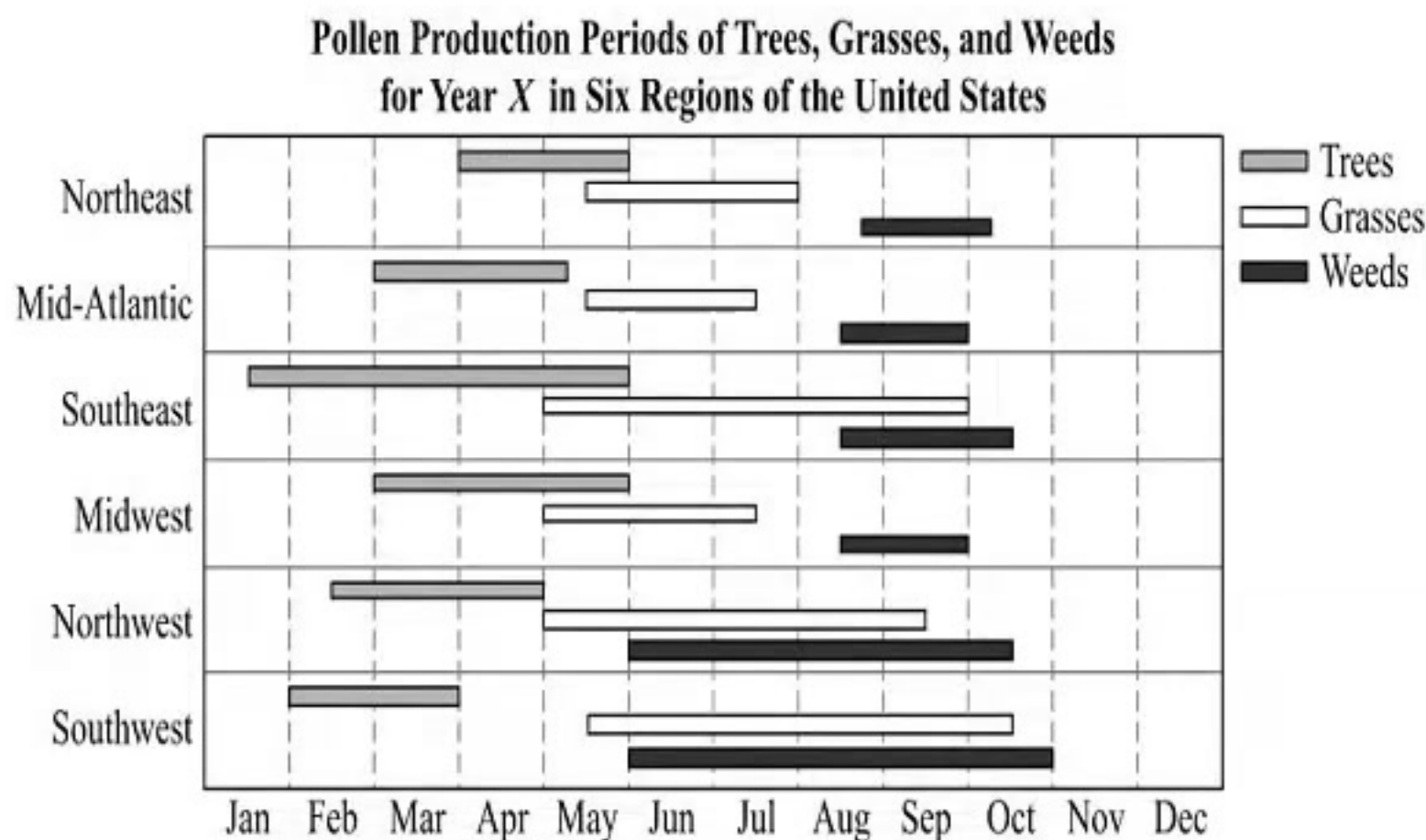
Which of the following is closest to the median length of the time periods, in months, during which trees produced pollen in the six regions?

☐ 2.0☐ 2.4☐ 2.8☐ 3.0☐ 3.2

Select one answer choice.



Questions 6–8 are based on the following data.



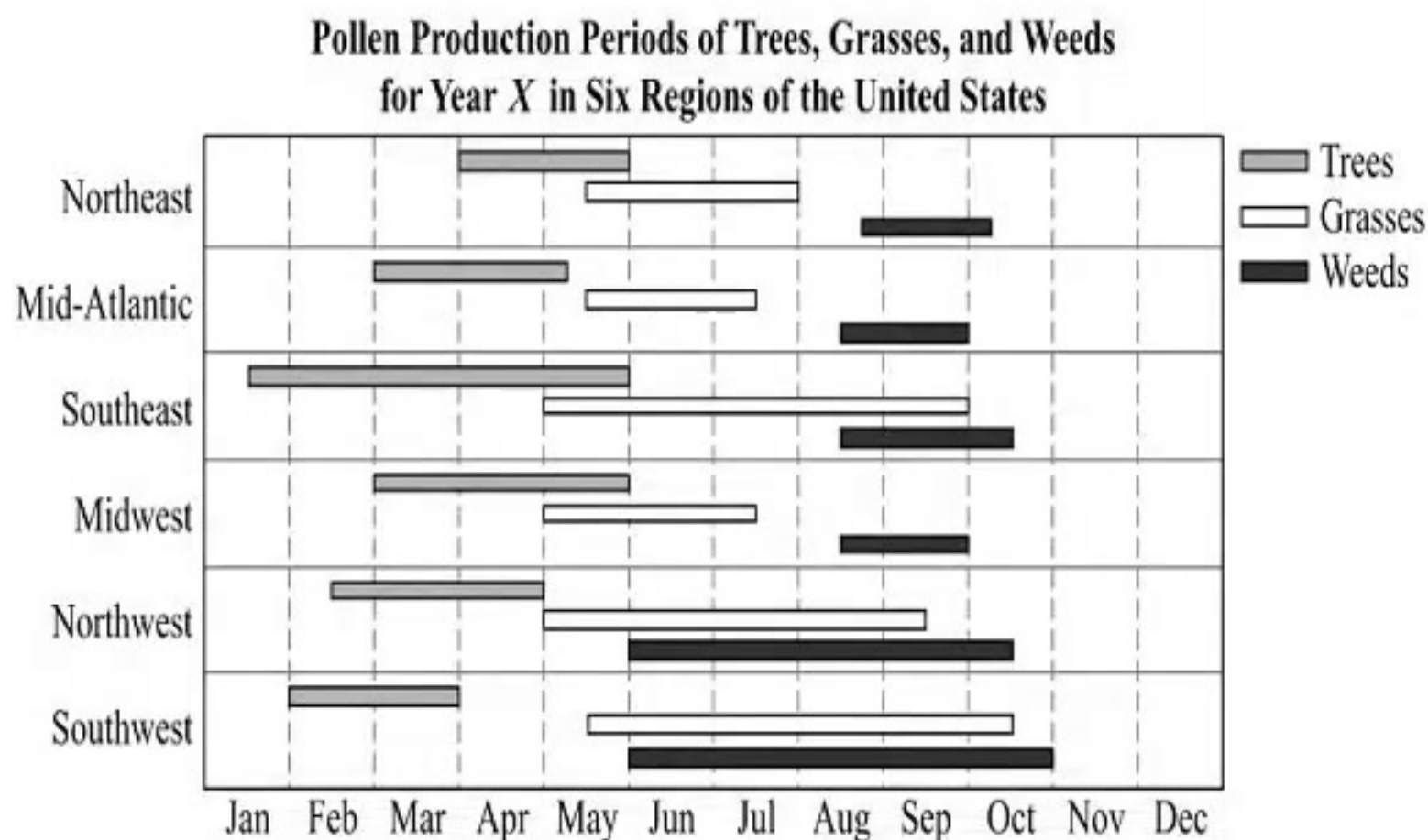
Which of the following is closest to the median length of the time periods, in months, during which trees produced pollen in the six regions?

☐ 2.0☐ 2.4☐ 2.8☐ 3.0☐ 3.2

Select one answer choice.



Questions 6–8 are based on the following data.



For approximately what fraction of Year X was pollen produced by both grasses and weeds in the Southwest?

☐ $\frac{9}{24}$

☐ $\frac{11}{24}$

☐ $\frac{12}{24}$

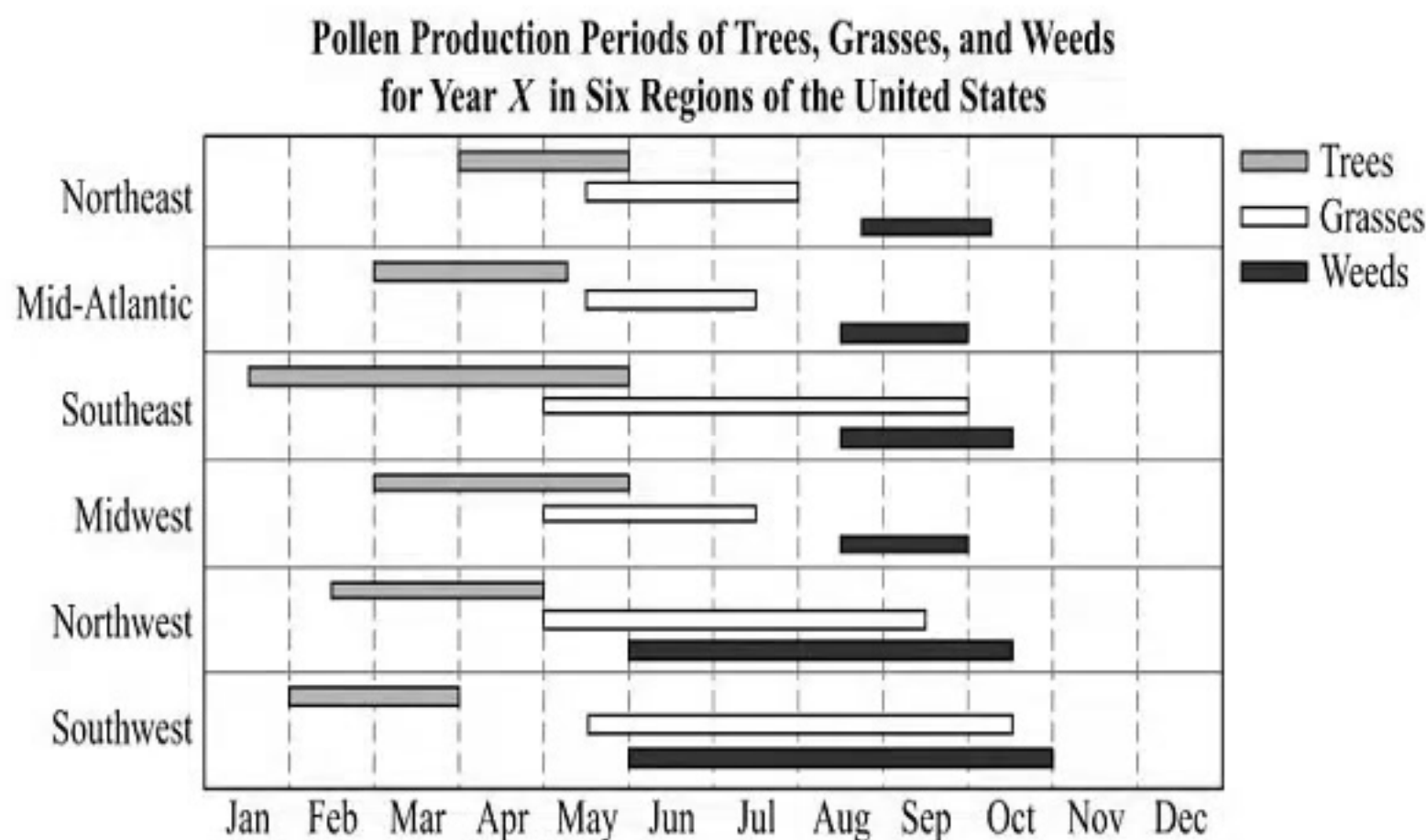
☐ $\frac{14}{24}$

☐ $\frac{20}{24}$

Select one answer choice.



Questions 6–8 are based on the following data.



For year X , which of the following is closest to the total length, in months, of the time period during which no pollen was produced by trees or grasses in any of the six regions?

☐ 0.5☐ 1.0☐ 2.0☐ 2.5☐ 3.0

Select one answer choice.



A square field in Town A measures 100 meters on each side. A square field in Town B has an area 4 times the area of the field in Town A . What is the perimeter, in meters, of the square field in Town B ?

- ☐ 200
- ☐ 400
- ☐ 600
- ☐ 800
- ☐ 1,600

Select one answer choice.



If $x > 1$, then $\frac{1}{x} - \frac{1}{x}(1 - x) =$

☐ $-(1 - x)$

☐ $1 - x$

☐ $1 + x$

☐ 0

☐ 1

Select one answer choice.



If $x > 1$, then $\frac{1}{x} - \frac{1}{x}(1 - x) =$

☐ $-(1 - x)$

☐ $1 - x$

☐ $1 + x$

☐ 0

☐ 1

Select one answer choice.



A game involves 52 playing chips, which are either blue, red, or white. The number of blue chips is $\frac{1}{2}$ the number of red chips, and the number of white chips is $\frac{2}{3}$ the number of red chips. Which of the following statements are true?

Indicate all such statements.

- ☐ The number of blue chips is $\frac{3}{4}$ the number of white chips.
- ☐ The number of white chips is 4 more than the number of blue chips.
- ☐ The number of red chips is 36.



The measures of the angles of a quadrilateral are in the ratio of 1 to 2 to 3 to 4. The smallest angle measures x° and the largest angle measures y° . What is the value of $y - x$?

 $y - x =$



Section 4 of 5

Verbal Reasoning

15 Questions

23 minutes (standard time)

For each question, indicate the best answer using the directions given. If you need more detailed directions, select **Help** at any time.

If a question has answer choices with **ovals** ☐, then the correct answer consists of a single choice. If a question has answer choices with **square boxes** ☐, then the correct answer consists of one or more answer choices. Read the directions for each question carefully. The directions will indicate if you should select one or more answer choices. To answer questions based on a reading passage, you may need to scroll to read the entire passage. You may also use your keyboard to navigate through the passage.

Select **Continue** to proceed.

Select one entry for the blank. Fill the blank in the way that best completes the text.

Although researchers have studied cartilage' s behavior on the macroscopic scale, the several forces that scientists believe to be active on the molecular level have never been _____.

disentangled
divulged
invoked
accepted
developed

Select one answer choice.

For each blank select one entry from the corresponding column of choices. Fill all blanks in the way that best completes the text.

Most of the bank-efficiency literature suggests that bank mergers culminating in larger, more complex organizations (i) _____ significant economies of scale or scope that are (ii) _____ to consumers. Indeed, some of this literature suggests diseconomies, which would lead to higher loan rates or lower deposit rates for consumers.

Blank (i)

fail to provide
seek to create
cannot but generate

Blank (ii)

beneficial
responsive
transparent

Select one entry from each column.

For each blank select one entry from the corresponding column of choices. Fill all blanks in the way that best completes the text.

Reviews written by music critic and composer Stephenson were hardly (i)_____: musicians who performed his music could count on sympathetic coverage, while those who ignored him were held to (ii)_____ standards.

Blank (i)

disinterested
lucid
conventional

Blank (ii)

exacting
minimal
accepted

Select one entry from each column.

For each blank select one entry from the corresponding column of choices. Fill all blanks in the way that best completes the text.

Every few years, as a reviewer, one encounters a novel whose (i) _____ are so many in number, and so (ii) _____, that to correct them fully would produce a text that exceeded the novel itself in length. Faced with such a book, one wishes only to let it slip quietly to the seabed of culture, there to join thousands of other (iii) _____ books in their slow disintegration.

Blank (i)

felicities
ineptitudes
curiosities

Blank (ii)

thoroughgoing
innocuous
delightful

Blank (iii)

profound
incompetent
indispensable

Select one entry from each column.

Questions 5 and 6 are based on this passage.

Many historians today stress the challenges that Queen Elizabeth I of England (reigned 1558–1603) faced as a female ruler in the patriarchal sixteenth century, but these difficulties are often overstated. Of course, early modern English society was deeply patriarchal: male primogeniture governed most property arrangements as well as the laws of royal succession; and in theory, at least, women were not expected to assert any independent authority and were deemed subservient to male relatives. Nevertheless, female rule was no great novelty in the sixteenth century, and at the time of Elizabeth's accession, barely a murmur was heard questioning her legitimacy. Indeed, the prevailing sentiment within England was not concern but rather relief that Elizabeth's succession was relatively smooth and essentially undisputed.

The highlighted sentence performs which of the following functions in the passage?

- ☐ It clarifies the position of historians discussed earlier in the passage.
- ☐ It provides a rationale for an exaggerated statement made earlier in the passage.
- ☐ It summarizes a claim made earlier in the passage.
- ☐ It suggests a reason that the claim in the following sentence might seem surprising.
- ☐ It offers a conclusion that might seem surprising in light of the previous sentence.

Select one answer choice.

Questions 5 and 6 are based on this passage.

Many historians today stress the challenges that Queen Elizabeth I of England (reigned 1558–1603) faced as a female ruler in the patriarchal sixteenth century, but these difficulties are often overstated. Of course, early modern English society was deeply patriarchal: male primogeniture governed most property arrangements as well as the laws of royal succession; and in theory, at least, women were not expected to assert any independent authority and were deemed subservient to male relatives. Nevertheless, female rule was no great novelty in the sixteenth century, and at the time of Elizabeth's accession, barely a murmur was heard questioning her legitimacy. Indeed, the prevailing sentiment within England was not concern but rather relief that Elizabeth's succession was relatively smooth and essentially undisputed.

Consider each of the choices separately and select all that apply.

Which of the following statements about Queen Elizabeth's succession to the throne is suggested by the passage?

- ☐ It placed Elizabeth in an unprecedented position for a woman at that time.
- ☐ It was not followed by as many difficulties as some modern-day historians have claimed.
- ☐ Its acceptance might seem surprising given the attitudes toward gender that prevailed at that time.

Select one or more answer choices.

Questions 7 to 9 are based on this passage.

In seventeenth-century Venice, the invention of the opera house as a public institution altered the relationship of the audience to the operatic performance on the stage and to each other. The stage performance was now separated from the audience both by the proscenium arch, which framed the action, and by the orchestra. The occupants of the boxes faced one another across the auditorium as well as the stage. Seeing and being seen was as much a function of the opera house as seeing and hearing what took place on the stage. American visitors to Italy were routinely shocked at the casual ways of the boxholders and at the little attention they seemed to pay to the opera. There was incessant talk in the boxes, banging of doors, a restless to-and-fro.

Public display was only half the story. The public nature of Venice's social life was balanced by a passion for disguise. Much of its political activity was conducted in secrecy. Masks were usually worn during carnival, attracting the crowned heads of Europe, who could play there incognito. The Venetian opera box was the equivalent of the mask. With its door shut, with shutters pulled across the front, the box became a place of privacy, an association that still exists in our time.

Consider each of the choices separately and select all that apply.

According to the passage, the boxes of public opera houses served to

- ☐ create privacy for their occupants
- ☐ frame the opera's action for the audience
- ☐ advertise the presence of their occupants

Select one or more answer choices.

Questions 7 to 9 are based on this passage.

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Regarding the "crowned heads of Europe," the author implies which of the following?

- ☐ Their presence in the opera audience influenced the kind of subject matter that operas presented.
- ☐ They thought of Venice as an advantageous location for clandestine politics and diplomacy.
- ☐ They took advantage of a Venetian tradition in order to amuse themselves.
- ☐ Their use of the opera box made it fashionable for the Venetian upper classes.
- ☐ They came to carnival to enjoy opera and the sight of masked revelers.

Select one answer choice.

Questions 7 to 9 are based on this passage.

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In the last paragraph, the author draws attention to the way that the opera box

- ☐ encouraged the creation of operas with hidden political content
- ☐ intensified opera's identification with the social elite
- ☐ led to the decline of the Venetian carnival
- ☐ formalized social stratification within the audience
- ☐ fitted independently identifiable tendencies in Venetian society

Select one answer choice.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

Despite her rather _____ choices, Moreland was neither a rebellious spirit nor someone who saw herself as anything out of the ordinary.

☐ unconventional

☐ impractical

☐ quirky

☐ flamboyant

☐ successful

☐ lucrative

Select two answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

The critic claims that, contrary to its reputation for _____, the novelist's prose is full of opaque language games.

☐ scrupulousness

☐ simplicity

☐ mendacity

☐ artlessness

☐ polish

☐ meticulousness

Select two answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

Although extreme and singular events occur in a black hole, its event horizon boundary is _____: space travelers crossing it would be subject to the same physical laws that apply anywhere else.

- ☐ illusory
- ☐ obscure
- ☐ unremarkable
- ☐ immutable
- ☐ unchanging
- ☐ commonplace

Select two answer choices.

Question 13 is based on this passage.

Because of strong overall stylistic similarities, prehistoric cave paintings in El Castillo, Spain, and others in Altamira, Spain, were thought to date from the same period. Recent pigment analyses, however, established the Altamira paintings as by far the older and as about the same age as a third group of paintings done in a different overall style. Nevertheless, this case does not prove that stylistic similarities cannot be a reliable indication of closeness in age, since _____.

Which of the following most logically completes the passage?

- ☐ experts who knew the results of the pigment analyses reexamined these paintings and found no previously unnoticed stylistic similarities between the Altamira paintings and the paintings in the third group
- ☐ there are certain stylistic features that the Altamira paintings share with all known paintings of their period and do not share with the El Castillo paintings
- ☐ before dating by pigment analysis was widely practiced, stylistic similarities were the only means of determining whether two prehistoric cave paintings dated from the same period
- ☐ determining the age of a prehistoric cave painting simply by reference to the age of the pigments used in that painting substitutes technology for artistic judgment
- ☐ there are experts who maintain that the results of the recent pigment analyses leave the striking stylistic similarities between the El Castillo paintings and the Altamira paintings unexplained

Select one answer choice.

Questions 14 and 15 are based on this passage.

Important historical differences between Europe and North America may explain why many European migratory songbirds, in contrast to North American migratory songbirds, thrive in forest habitats fragmented by human activity. First, the Pleistocene glaciations may have been more severe in Europe than in North America. Virtually all Europe's forests were erased during the most recent Ice Age. Any European songbirds dependent upon large, intact blocks of forest were unlikely to survive. In the southern part of North America, however, some forests managed to persist through the coldest times, to the advantage of some forest-dwelling songbirds. Second, European forests were being abused by humans for thousands of years before the North American forests were. Consequently, European birds have had more time to adapt to a human-dominated landscape.

According to the passage, which of the following is true about today's European migratory songbirds?

- ☐ They generally are not dependent on large, intact blocks of forest.
- ☐ They generally are less sensitive to cold temperatures than are North American songbirds.
- ☐ They tend to be unevenly distributed across Europe because of differences in habitat.
- ☐ Most would be unlikely to survive additional fragmentation of their preferred habitat.
- ☐ Most represent species that evolved after the Pleistocene glaciations.

Select one answer choice.

Questions 14 and 15 are based on this passage.

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In the author's argument, the observation made in the highlighted portion of the passage primarily serves to

- ☐ identify a factor that helps explain why Europe's songbird population is significantly smaller than North America's
- ☐ help explain a distinctive characteristic of European songbirds
- ☐ shed light on why there is less habitat available for songbirds in Europe than for songbirds in North America
- ☐ point to a reason why European songbirds have had more time to adapt to a human-dominated landscape than have North American birds
- ☐ emphasize the long duration of the Pleistocene glaciations in North America relative to the Pleistocene glaciations in Europe

Select one answer choice.



End of Section

You have reached the end of this section. You have time remaining to review. As long as there is time remaining, you can check your work. Once you leave this section, you WILL NOT be able to return to it.

Select **Review** to go back to the Review screen.

Select **Return** to go to the last question in this section.

Select **Continue** to proceed to the next section of the test.



Quantitative Reasoning

15 Questions

26 minutes (standard time)

For each question, indicate the best answer, using the directions given. If you need more detailed directions, select **Help** at any time.

An on-screen calculator is available for each question in this section. To use the calculator, select the calculator button.

If a question has answer choices with **ovals** ☐, then the correct answer consists of a single choice. If a question has answer choices with **square boxes** ☐, then the correct answer consists of one or more answer choices. Read the directions for each question carefully. The directions will indicate if you should select one or more answer choices. To answer questions based on a data presentation, you may need to scroll or use your keyboard to access the entire presentation.

All numbers used are real numbers.

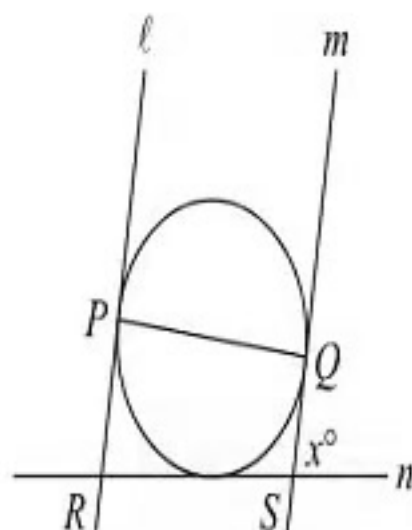
All figures are assumed to lie in a plane unless otherwise indicated.

Geometric figures, such as lines, circles, triangles, and quadrilaterals, **are not necessarily** drawn to scale. That is, you should **not** assume that quantities such as lengths and angle measures are as they appear in a figure. You should assume, however, that lines shown as straight are actually straight, points on a line are in the order shown, and more generally, all geometric objects are in the relative positions shown. For questions with geometric figures, you should base your answers on geometric reasoning, not on estimating or comparing quantities by sight or by measurement.

Coordinate systems, such as xy -planes and number lines, are drawn to scale; therefore you can read, estimate, or compare quantities in such figures by sight or by measurement.

Graphical data presentations, such as bar graphs, circle graphs, and line graphs, are drawn to scale; therefore, you can read, estimate, or compare data values by sight or by measurement.

Select **Continue** to proceed.



PQ is a diameter of the circle, line ℓ is tangent to the circle at P , line m is tangent to the circle at Q , line n is tangent to the circle, and $x < 90$.

Quantity A

RS

Quantity B

PQ

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.

Select one answer choice.



n is a positive odd integer.

$$a = \frac{n+1}{2}$$

$$b = \frac{n-1}{2}$$

Quantity A

$$(-2)^a$$

Quantity B

$$(-2)^b$$

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.



Number of Hours (h) Spent Exercising	Percent of Club Members
$0 \leq h < 1$	35%
$1 \leq h < 2$	30%
$2 \leq h < 3$	20%
$3 \leq h$	15%

The table shows the percent distribution of the 1,500 members of a sports club according to the amount of time they spent exercising yesterday.

Quantity A

The number of club members who exercised between 1 hour and 1 hour 40 minutes yesterday

Quantity B

300

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.



For any number x , the value of the function $f(x)$ equals the greatest integer less than or equal to x .

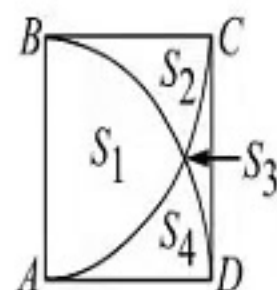
Quantity A

$$f(-1.5)$$

Quantity B

$$f(-2)$$

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.



Square $ABCD$ has sides of length 1. Arc AC is an arc of the circle of radius 1 centered at B . Arc BD is an arc of the circle of radius 1 centered at A . These arcs divide the square into four regions, S_1 , S_2 , S_3 , and S_4 .

Quantity A

The area of region S_1 minus the area of region S_3

Quantity B

$$\frac{\pi}{2} - 1$$

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.



In a survey conducted by television industry analysts, program X had an average rating last year of 12.5 points while program Y had an average rating of 12.1 points. If each point on the rating scale represented 930,000 households that viewed the program, how many more households on average viewed program X than viewed program Y ?

- ☐ 37,200
- ☐ 55,800
- ☐ 372,000
- ☐ 558,000
- ☐ 3,720,000



If w , x , y , and z are positive numbers such that $x = 2w$, $y = 3x$, and $z = 4y$, which of the following statements is true about the four sums $w + y$, $w + z$, $x + y$, and $x + z$?

- ☐ $w + y < w + z < x + y < x + z$
- ☐ $w + y < w + z < x + z < x + y$
- ☐ $w + y < x + y < w + z < x + z$
- ☐ $x + y < w + y < w + z < x + z$
- ☐ $x + y < w + y < x + z < w + z$

Select one answer choice.



For each of the last 5 years, the population of a colony of beetles increased by 8 percent of the preceding year's population. If P represents the current population of the colony, which of the following best represents the population 5 years ago, in terms of P ?

☐ $(5)(1.08P^{-1})$

☐ $(1.08)^{-5}P^{-1}$

☐ $(1.08P)^{-5}$

☐ $(1.08)^{-5}P$

☐ $(1.08)^{-5}P^5$



From a set of 100 numbers, half were selected to form group I, and 60 percent of the remaining numbers were selected to form group II. The average (arithmetic mean) of the numbers in group I is 24.4, and the average of the numbers in group II is 31.5. Which of the following is closest to the average of the numbers in groups I and II combined?

☐ 27.1

☐ 27.6

☐ 27.8

☐ 28.0

☐ 28.3

Select one answer choice.



In the xy -plane, circle C has center $(0, 0)$ and radius r , where r is a positive integer. If the points $(10, 6)$, $(9, 9)$, and $(8, 11)$ all lie inside circle C , what is the least possible value of r ?

- ☐ 13
- ☐ 14
- ☐ 16
- ☐ 18
- ☐ 19



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- ☐ 13
- ☐ 14
- ☐ 16
- ☐ 18
- ☐ 19



If n is a positive integer, which of the following CANNOT be the units digit of $2^n - 1$?

☐ 1

☐ 3

☐ 5

☐ 7

☐ 9

Select one answer choice.



Consider the quadratic equation $x^2 + bx + c = 0$, where b and c are constants. If $x = -2$ and $x = 3$ are the two solutions of the equation, what is the value of c ?

$c =$

**Children's Access to Computers, 1984**

Access to Computer	Percent
At home	15
At school	42
At home and at school	10

According to the data in the table, what percent of the children had access to a computer either at home, at school, or both?

- ☐ 67%
- ☐ 57%
- ☐ 52%
- ☐ 47%
- ☐ 42%

Select one answer choice.



Among 16 textbooks purchased by a bookstore, each textbook was either a mathematics textbook or a chemistry textbook, and each textbook was either new or used. Of the mathematics textbooks purchased, $\frac{1}{2}$ were used. Of the chemistry textbooks purchased, $\frac{2}{3}$ were used. Which of the following could be the total number of textbooks purchased that were used?

Indicate all such numbers.

- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 9
- ☐ 10
- ☐ 11
- ☐ 12



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Indicate all such numbers.

- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 9
- ☐ 10
- ☐ 11
- ☐ 12



Two balls are to be randomly selected from a bag, one at a time and without replacement. The probability that the first ball selected will be red is $\frac{5}{8}$. If the first ball selected is not red, the probability that the second ball selected will be red is $\frac{2}{3}$. What is the probability that the first or the second ball selected will be red?

Give your answer as a fraction.

Enter your answer as a fraction, with the numerator and denominator in their respective answer boxes. Backspace to erase.



End of Section

You have reached the end of this section. You have time remaining to review. As long as there is time remaining, you can check your work. Once you leave this section, you **WILL NOT** be able to return to it.

Select **Review** to go back to the Review screen.

Select **Return** to go to the last question in this section.

Select **Continue** to proceed to the next section of the test.



Report or Cancel Your Scores

You now have the option to either report or cancel your GRE General Test scores from this testing session.

If you choose to report your scores from this testing session, they will become a part of your cumulative record.

If you choose to cancel your scores from this testing session, **scores will not be reported for any section of this test.**

Select **Report Scores** or **Cancel Scores** now.



Report Your Scores

You have chosen to REPORT your GRE General Test scores, or you did not respond on the previous screen.

This is the last chance to change your mind.

Once presented, scores CANNOT be canceled.

To confirm that you would like to report your scores, select **Report Scores**. To cancel your scores, select **Cancel Scores**.

Select Your Score Recipients

If you know where you want to send your test scores, you can select up to four (4) institutions now as part of your test fee. You can choose to send scores from your most recent test administration or from all your test administrations. Select the institutions by clicking "Add a Score Recipient" below.

If you prefer, you can select score recipients at a later date for a fee through your ETS Account. At that time, you can choose to send scores from your most recent test administration, from all your test administrations, or from any specific test administration(s).

Selected Score Recipient(s)

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Please select **Continue** to exit.

V1

1.B 2.CD 3.BFH 4.C 5.AB 6.CF 7.AC 8.CF 9.DF 10.E 11. D 12.C

V2

1.A 2.AD 3.AD 4.BDH 5.E 6.BC 7.AC 8.C 9.E 10.AC 11.BD 12.CF 13.B 14.A 15.B